CPRE 5360: Term Paper Topic Submission
Captain Scott Russell: Fall 2026

Background Knowledge

Prior to CPRE 5360 I have had both academic and operational experience in digital forensics.

Academically, I completed CPRE 4360 (Computer and Network Forensics) with Dr. Julie Rursch last semester, which covered the forensic process end to end. This included evidence identification and acquisition, chain of custody and legal admissibility, file system and disk analysis, data carving and recovery of deleted artifacts, network traffic analysis, and the reporting standards expected of an examiner whose work may be challenged in court.

Professionally, I hold the GIAC Certified Forensic Analyst (GCFA) certification, earned through SANS FOR508: Advanced Incident Response, Threat Hunting, and Digital Forensics. FOR508 is built around enterprise scale intrusion investigation rather than single device examination, and is a mile wide and an inch deep.

My Air Force Specialty Code is 17SA, Offensive Cyberspace Operations Officer. I have been a cyber officer for the past 7-years. Prior to my current assignment as an instructor at Air Force ROTC Detachment 250 at Iowa State, I was a qualified Offensive Cyber Operator at Joint Base San Antonio, Lackland, Texas, supporting United States Cyber Command and the National Security Agency. I served on the "Access Management Team" with the 367th Cyberspace Operations Squadron conducting operational support for USCYBERCOM and joint partners conducting missions at the Joint Mission Operations Center in Texas (JMOC-T).

That offensive background is directly relevant here. Forensics is fundamentally an adversarial discipline, and my operational experience gives me a working understanding of how an attacker reasons about the artifacts they leave behind, what they attempt to suppress, and where defensive tooling tends to be blind. I want a topic where I can evaluate a forensic capability against a realistically evasive adversary trying to hide information on a system.

Proposed Topic: Cross-Checking Windows Artifacts to Detect Anti-Forensic Tampering

My planned topic falls under ***forensic tool validation*** and ***triage tool analysis*** from the potential topic list in the assignment description.

Windows keeps several independent records of what has happened on a system: the Master File Table (MFT), the change journal (UsnJrnl:$J), link files, execution artifacts such as Prefetch, and the event logs. They are written by different parts of the operating system for different reasons, and none was originally designed as evidence. Someone covering their tracks can alter one or two of them, but keeping all of them consistent with each other is much harder, so a cover-up tends to leave the records disagreeing.

Existing tools mostly leave these checks to the examiner. Commercial suites such as EnCase and FTK, which we are going to use in this course, present the artifacts but leave the analyst to notice that two records disagree and decide whether it matters. More specialized tools automate an individual check, but each sign alone is a weak indicator, since logs may be overwritten in normal use. My question is

whether measuring how much several records disagree identifies tampering more reliably than any single check does.

There is relevant prior work to build on that I did some initial research on while creating this proposal. Palmbach and Breitinger [1] showed that the change journal, link files, Prefetch and event logs all carry evidence of timestamp manipulation, concluded that none of these sources is reliable on its own, and proposed five specific consistency rules across those artifacts. Vanini, Hargreaves, and Breitinger [2] proposed a scoring framework for how resistant each source is to tampering in the first place. Palmbach and Breitinger recommend combining artifacts to improve the odds of detection but do not measure whether combining actually helps, and their experiment used four test files on an otherwise unused system, so false positives on normal activity were never quantified. Vanini, Hargreaves, and Breitinger score how hard a source is to tamper with rather than whether a given system shows signs that tampering occurred. That measurement gap is where I want to work.

My rough plan is to build a small tool, likely in Python given my background, that parses several of these artifacts and produces a combined "disagreement score", starting from the five rules in [1] and extending them. I would test it on a lab virtual machine that I use normally so it accumulates ordinary activity, then apply a known set of concealment actions at times I record, so the correct answer is known in advance. I would then compare my score against the single-artifact checks using detection and false positive rates. Time permitting, I would also like to run the tool against the forensic images used in the projects of this course, as an additional test against data I did not build myself.

Between now and the November draft I intend to settle on what concealment techniques to draw from, survey the existing parsers so I am not rebuilding what already works, and review recent work on artifact inconsistency to confirm this work is novel in some way.

Scope will stay narrow: one Windows version, New Technology File System (NTFS)only, a short and stated list of concealment actions, and test systems I build rather than real evidence. I am well placed to choose those actions realistically, since deciding what traces to suppress was part of my operational work.

References (IEEE Style with links in-line)

[1] D. Palmbach and F. Breitinger, "Artifacts for detecting timestamp manipulation in NTFS on Windows and their reliability," *Forensic Science International: Digital Investigation*, vol. 32, 300920, 2020.

[2] C. Vanini, C. Hargreaves, and F. Breitinger, "Evaluating tamper resistance of digital forensic artifacts during event reconstruction," arXiv preprint arXiv: 2412.12814, 2024.